

Series and Integral Transforms

Theorems & Formulas Reference Sheet

Course 21183 — HIT, Prof. Michael Kroyter

Contents

1	Numerical Series	2
1.1	Basic Definitions	2
1.2	Convergence Tests	2
1.3	Absolute vs. Conditional Convergence	2
1.4	p -Series	2
2	Power Series	3
2.1	Radius of Convergence	3
2.2	Taylor / Maclaurin Series	3
2.3	Common Maclaurin Series	3
2.4	Taylor's Theorem with Remainder	3
2.5	Computing Limits via Taylor Series	4
3	Uniform Convergence	4
3.1	Sequences of Functions	4
3.2	Properties Preserved by Uniform Convergence	5
3.3	Series of Functions	5
4	Fourier Series	6
4.1	Definition	6
4.2	Dirichlet Conditions	6
4.3	Even and Odd Functions	6
4.4	Parseval's Identity	6
4.5	Complex Form	6
5	Fourier Transform	6
5.1	Definition	6
5.2	Key Properties	7
5.3	Parseval / Plancherel Theorem	7
6	Laplace Transform	7
6.1	Definition	7
6.2	Common Transforms	7
6.3	Convolution Theorem	7

1 Numerical Series

1.1 Basic Definitions

- The series $\sum_{n=1}^{\infty} a_n$ **converges** if the partial sums $S_N = \sum_{n=1}^N a_n$ have a finite limit.
- **Necessary condition:** if $\sum a_n$ converges, then $a_n \rightarrow 0$.
Contrapositive: if $a_n \not\rightarrow 0$, the series diverges.

1.2 Convergence Tests

Test	Statement
Comparison	If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, so does $\sum a_n$.
Limit Comparison	If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \in (0, \infty)$, both series behave the same.
Ratio (D'Alembert)	$L = \lim \left \frac{a_{n+1}}{a_n} \right $: converges if $L < 1$, diverges if $L > 1$, inconclusive if $L = 1$.
Root (Cauchy)	$L = \limsup a_n ^{1/n}$: converges if $L < 1$, diverges if $L > 1$.
Integral	$\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x) dx$ converge/diverge together (for f positive and decreasing).
Alternating (Leibniz)	$\sum (-1)^n b_n$ converges if $b_n \geq 0$ and $b_n \searrow 0$.
Dirichlet	$\sum a_n b_n$ converges if partial sums of a_n are bounded and $b_n \searrow 0$.
Abel	$\sum a_n b_n$ converges if $\sum a_n$ converges and b_n is monotone and bounded.

1.3 Absolute vs. Conditional Convergence

- **Absolutely convergent:** $\sum |a_n| < \infty$ (implies convergent).
- **Conditionally convergent:** $\sum a_n$ converges but $\sum |a_n|$ diverges.

1.4 p -Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges if and only if } p > 1.$$

2 Power Series

2.1 Radius of Convergence

The series $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges absolutely for $|x-a| < R$, where

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}} \quad (\text{Cauchy-Hadamard formula})$$

Also: $R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ when the limit exists.

2.2 Taylor / Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

2.3 Common Maclaurin Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad R = \infty$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad R = \infty$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, \quad |x| \leq 1, x \neq -1$$

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \quad |x| < 1$$

2.4 Taylor's Theorem with Remainder

Theorem 2.1 (Taylor's Theorem — Lagrange Remainder). *If f has $n+1$ continuous derivatives on an interval containing a , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x),$$

where the **Lagrange remainder** is

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$$

for some c strictly between a and x .

Error bound:

$$|R_n(x)| \leq \frac{M_{n+1}}{(n+1)!} |x-a|^{n+1}, \quad M_{n+1} = \sup_{t \text{ between } a \text{ and } x} |f^{(n+1)}(t)|.$$

Workflow for “how many terms” questions: Bound $|R_n(x)|$ using M_{n+1} , then solve $\frac{M_{n+1}}{(n+1)!} |x-a|^{n+1} < \varepsilon$ for the smallest n .

2.5 Computing Limits via Taylor Series

To evaluate $\lim_{x \rightarrow 0} g(x)$:

1. Expand numerator and denominator as power series around 0.
2. Cancel common factors of x^k .
3. Read off the limit from the leading terms.

For $\lim_{x \rightarrow 0} \frac{d^k}{dx^k} [f(x)]$: write $f(x) = \sum a_n x^n$, then $f^{(k)}(0) = k! a_k$.

3 Uniform Convergence

3.1 Sequences of Functions

Definition 3.1. Let $f_n, f : D \rightarrow \mathbb{R}$.

- **Pointwise convergence:** $f_n \rightarrow f$ if for each $x \in D$, $f_n(x) \rightarrow f(x)$.
- **Uniform convergence** ($f_n \Rightarrow f$):

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0.$$

Uniform $\Rightarrow$ pointwise, but pointwise $\not\Rightarrow$ uniform in general.

Standard method: Compute $M_n = \sup_{x \in D} |f_n(x) - f(x)|$ (find the max by differentiation or AM-GM). If $M_n \rightarrow 0$: uniform; if $M_n \not\rightarrow 0$: not uniform.

Negation: $f_n \not\Rightarrow f$ iff $\exists \varepsilon > 0$ and a sequence $x_n \in D$ with $|f_n(x_n) - f(x_n)| \geq \varepsilon$. A typical choice is $x_n = \arg \max_x |f_n(x) - f(x)|$.

Example 3.2 ($f_n(x) = x^n$ on $[0, 1]$). Pointwise limit: $f(x) = 0$. But $\sup_{x \in [0, 1]} |x^n - 0| = 1 \not\rightarrow 0$. Not uniform. On $[0, r]$ for any $r < 1$: $\sup = r^n \rightarrow 0$. Uniform.

3.2 Properties Preserved by Uniform Convergence

Theorem 3.3 (Continuity). *If each f_n is continuous on D and $f_n \Rightarrow f$, then f is continuous on D .*

Theorem 3.4 (Term-by-term Integration). *If $f_n \Rightarrow f$ on $[a, b]$, then*

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx.$$

Theorem 3.5 (Term-by-term Differentiation). *If $f_n \rightarrow f$ pointwise on $[a, b]$, each f_n is C^1 , and $f'_n \Rightarrow g$ uniformly, then f is differentiable and $f' = g$.*

Remark. Counterexample for pointwise-only: $f_n(x) = n^2 x e^{-nx}$ on $[0, 1]$ converges pointwise to 0, but $\int_0^1 f_n dx = 1 \not\rightarrow 0$. The convergence is not uniform.

3.3 Series of Functions

A series $\sum_{n=1}^{\infty} u_n(x)$ **converges uniformly** on D if its partial sums $S_N(x) = \sum_{n=1}^N u_n(x)$ converge uniformly on D .

Theorem 3.6 (Weierstrass M -Test). *If $|u_n(x)| \leq M_n$ for all $x \in D$ and $\sum_{n=1}^{\infty} M_n < \infty$, then $\sum u_n(x)$ converges uniformly and absolutely on D .*

Theorem 3.7 (Dirichlet's Test — Uniform Version). *$\sum a_n(x) b_n(x)$ converges uniformly on D if:*

1. *The partial sums $A_N(x) = \sum_{n=1}^N a_n(x)$ are uniformly bounded on D .*
2. *$b_n(x) \searrow 0$ monotonically and uniformly on D .*

Theorem 3.8 (Abel's Test — Uniform Version). *$\sum a_n(x) b_n(x)$ converges uniformly on D if:*

1. *$\sum a_n(x)$ converges uniformly on D .*
2. *$b_n(x)$ is monotone and uniformly bounded on D .*

Key fact about power series: $\sum_{n=0}^{\infty} c_n x^n$ with radius of convergence R converges uniformly on $[-r, r]$ for any $r < R$, but generally *not* uniformly on the open interval $(-R, R)$.

Properties (same as for sequences): if $\sum u_n$ converges uniformly and each u_n is continuous, then the sum is continuous; term-by-term integration and differentiation are valid under uniform convergence (for differentiation: need $\sum u'_n$ to converge uniformly).

4 Fourier Series

4.1 Definition

For f periodic with period $2L$:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Coefficients:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

4.2 Dirichlet Conditions

If f is piecewise smooth and periodic, the Fourier series converges to:

- $f(x)$ at points of continuity,
- $\frac{f(x^+) + f(x^-)}{2}$ at jump discontinuities.

4.3 Even and Odd Functions

- **Even** f : $b_n = 0$ (cosine series only).
- **Odd** f : $a_n = 0$ (sine series only).

4.4 Parseval's Identity

$$\frac{1}{L} \int_{-L}^L |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

4.5 Complex Form

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

5 Fourier Transform

5.1 Definition

$$\hat{f}(\xi) = \mathcal{F}\{f\}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx$$

(Alternative convention: $\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$)

5.2 Key Properties

Property	Formula
Linearity	$\mathcal{F}\{af + bg\} = a\hat{f} + b\hat{g}$
Shift	$\mathcal{F}\{f(x - a)\} = e^{-2\pi ia\xi} \hat{f}(\xi)$
Scaling	$\mathcal{F}\{f(ax)\} = \frac{1}{ a } \hat{f}(\xi/a)$
Derivative	$\mathcal{F}\{f'\}(\xi) = 2\pi i\xi \hat{f}(\xi)$
Convolution	$\mathcal{F}\{f * g\} = \hat{f} \cdot \hat{g}$

5.3 Parseval / Plancherel Theorem

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$$

6 Laplace Transform

6.1 Definition

$$\mathcal{L}\{f\}(s) = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

6.2 Common Transforms

$f(t)$	$F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(bt)$	$\frac{b}{s^2 + b^2}$
$\cos(bt)$	$\frac{s}{s^2 + b^2}$
$e^{at}f(t)$	$F(s - a)$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$

6.3 Convolution Theorem

$$\mathcal{L}\{f * g\} = F(s) \cdot G(s)$$